学习理论
Learning Theory

明玉瑞 Yurui Ming

yrming@gmail.com

声明
Disclaimer

- 本讲义在准备过程中由于时间所限，所用材料来源并未规范标示引用来源。所引材料仅用于教学所用，作者无意侵犯原著者之知识版权，所引材料之知识版权均归原著者所有；若原著者介意之，请联系作者更正及删除。

  These slides have been developed free-of-charge for educational purposes only and without any commercial interest. However, time limit during the preparation incurs the situation that not all the sources of the materials (texts or images) are properly referenced or clearly manifested. But all contents are solely for teaching and the author is with no intention to infringe the copyright bestowed on the originators and all credits go to corresponding IP holders. Please address the author have you had any concern so he can act on a remedy including deletion per request.

算法学习的目标
Goal of Learning Algorithms

- 早期的学习算法旨在实现对数据的精准拟合，例如，若一个分类器能对训练数据做出正确分类，则称其为一致性分类器。后来，人们意识到，算法的目的是要预测未来的数据，而非拟合手中已有的数据，即判定已知的结果。因此，有了泛化的概念，例如，分类器对训练集外的数据进行正确分类的能力，被称为其泛化能力。

  The early learning algorithms were designed to find such an accurate fit to the data. For example, a classifier is said to be consistent if it performed the correct classification of the training data. Later, people realize that the purpose of learning algorithm is to predict the real future instead of fitting the data in hand or predict the already know results. Therefore, the concept of generalization was introduce. For instance, The ability of a classifier to correctly classify data not in the training set is known as its generalization.

- 为了从理论上讨论算法学习的目标，我们引入一些模型。

  We introduce some models to discuss the goal of learning algorithms from theoretic perspective.

大概率近似正确学习模型
Probably Approximately Correct Learning (pac) Model

- 我们第一个要介绍的模型是大概率近似正确学习模型，它有如下关键假设：训练数据与测试数据均依据固定但未知的独立同分布（i.i.d.）$\mathcal{D}$生成。

  The first model introduced is called Probably Approximately Correct Learning (pac) model. The key assumption of it is that training and testing data are generated i.i.d. according to an fixed but unknown distribution $\mathcal{D}$.

- 当评估假设空间$\mathcal{H}$中某一假设（分类函数）h的“质量”时，需纳入未知分布$\mathcal{D}$的影响（即考虑假设空间$\mathcal{H}$中假设h产生的平均误差或期望误差），我们将这一度量称为风险泛函，如下所示：

  When we evaluate the “quality” of a hypothesis (classification function) $h \in \mathcal{H}$, we should take the unknown distribution $\mathcal{D}$ into account (i.e. “average error” or “expected error” made by the $h \in \mathcal{H}$). We call such measure risk functional and denote it as in the following:

$$\underset{\mathcal{D}}{\operatorname{err}}(h) = \mathcal{D}\{(x, y) \in X \times \{1, -1\} | h(x) \neq y\}$$

大概率近似正确学习模型
Probably Approximately Correct Learning (pac) Model

- 我们进一步展开，设$S=\{(x^1,y^1),\cdots,(x^l,y^l)\}$为依据分布$\mathcal{D}$独立同分布选取的含$l$个训练样本的集合。然后，将泛化误差$\underset{\mathcal{D}}{\mathrm{err}}(h_S)$视为随机变量，其取值由训练集$S$的随机选取过程决定。

  Further, let $S=\{(x^1,y^1),\cdots,(x^l,y^l)\}$ be a set of l training examples chosen i.i.d. according to $\mathcal{D}$, then treat the generalization error $\underset{\mathcal{D}}{\mathrm{err}}(h_S)$ as a r.v. depending on the random selection of S.

- 我们的目标是为该随机变量的分布波动寻找形式为$\varepsilon=\varepsilon(l,H,\delta)$的误差界。$\varepsilon=\varepsilon(l,H,\delta)$为关于样本量$l$、假设空间$H$和参数$\delta$的函数，其中$1-\delta$是由学习者指定的该误差界对应的置信水平。一般说来，当δ给定时，ε的上下界预示着泛化误差的最差与最佳水平。假设空间中的假设h_S所产生的误差，将小于与未知分布$\mathcal{D}$无关的误差界$\varepsilon(l,H,\delta)$。

  Our target is to find a bound of the trail of the distribution of r.v. in the form $\varepsilon=\varepsilon(l,H,\delta)$, where $\varepsilon(l,H,\delta)$ is a function of l, H and δ, where $1-\delta$ is the confidence level of the error bound which is given by learner. Generally speaking, for given δ, the upper bound and lower bound indicate the worst case and best case of generalization error of pac model. The error made by the hypothesis h_S will be less than the error bound $\varepsilon(l,H,\delta)$ that is not depend on the unknown distribution $\mathcal{D}$.

最小期望风险
Minimum Expected Risk

- 上面所讲的$\varepsilon(l,H,\delta)$并不好计算，在实际中，更常用的是最小期望风险。设$S=\{(x^1,y^1),\cdots,(x^l,y^l)\}\subseteq X\times\{-1,1\}$为依据概率密度函数$p(x,y)$的分布$\mathcal{D}$按独立同分布选取的训练样本的集合。假设空间$H$中的假设$h$产生的期望误分类误差为：

  The $\varepsilon(l,H,\delta)$ mentioned above is not easy to calculate, in practice, the minimum expected risk is more commonly-used. Let $S=\{(x^1,y^1),\cdots,(x^l,y^l)\}$ be a set of training examples chosen i.i.d. according to $\mathcal{D}$ with the probability density $p(x,y)$, the expected misclassification error made by $h\in H$ is:

$$R[h]=\int_{X\times\{-1,1\}}\frac{1}{2}|h(x)-y|\,dp(x,y)$$

- 最优假设h_{opt}^*应具有最小的期望风险，即对任意的$h\in H$，$R[h_{\text{opt}}^*]\le R[h]$。

  The ideal hypothesis h_{opt}^* should has the smallest expected risk $R[h_{\text{opt}}^*]\le R[h]$, $\forall\ h\in H$.

最小期望风险
Minimum Expected Risk

- $R[h]$的表达式似乎看起来更加光明，但由于$\mathcal{D}$是未知的，即$p(x,y)$是未知的，因此，并不能显式地按积分的方式计算$R[h]$。实际中，我们往往是将基于联合分布$p(x,y)$的期望风险替换为对训练样本的均值，从而得到经验风险：

  Seems the expression of $R[h]$ is more promising, however, since $\mathcal{D}$ is unknown, aka., $p(x,y)$ is unknown, so calculating $R[h]$ by integration is intractable. In practice, we usually replace the expected risk over $p(x,y)$ by an average over the training examples to get the empirical risk:

$$R_{\text{emp}}[h] = \frac{1}{l}\sum_{i=1}^{l}\frac{1}{2}\left|h\left(x^i\right) - y_i\right|$$

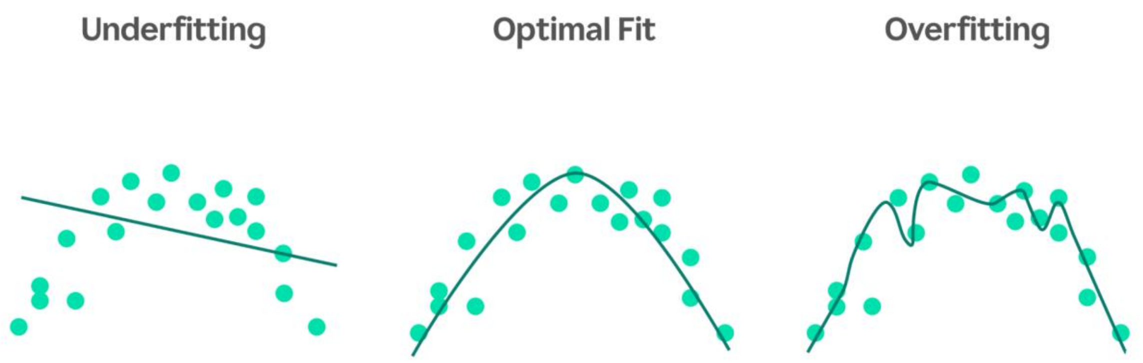
- 因此，现在我们的目标是求解具有最小经验风险的假设h_{emp}^*，即对任意的$h \in H$，$R_{\text{emp}}\left[h_{\text{emp}}^*\right] \le R_{\text{emp}}[h]$。需要指出的是，只关注经验风险会导致过拟合。

  Now our purpose is to find the hypothesis h_{emp}^* with the smallest empirical risk $R_{\text{emp}}\left[h_{\text{emp}}^*\right] \le R_{\text{emp}}[h], \forall\ h \in H$. Notably, only focusing on empirical risk will cause overfitting.

偏差–方差权衡
Bias/Variance Tradeoff

- 为了更好地了解过拟合的概念，我们从偏差与方差权衡的角度进行讨论。以下图为例，我们可以选择一个简单的多项式模型（例如$y=\theta_0+\theta_1 x$）来拟合，还可以选择更为复杂的多项式模型（例如$y=\theta_0+\theta_1 x+\theta_2 x^2+\cdots+\theta_5 x^5$）来拟合。

  To better explain the concept of overfitting, we discuss it from the perspective of bias/variance trade-off. As illustrated below, to fit the following data points, we can choose a simple polynomial model (such as $y=\theta_0+\theta_1 x$) or a complicated one (such as $y=\theta_0+\theta_1 x+\theta_2 x^2+\cdots+\theta_5 x^5$).



偏差–方差权衡
Bias/Variance Tradeoff

- 实际上，对数据采用5阶多项式（最右侧图）去拟合，并不能得到一个好模型。举例来说，尽管5阶多项式在训练集样本中，能从x（假定为房屋面积）很好地预测y（这里为房价），但我们并不认为这个模型能很好地预测不在训练集中的房屋价格。换句话说，从训练集中学到的规律泛化能力很差，即我们在上面讲的该假设/模型的泛化误差很大。读者不难注意到，上图中最左侧和最右侧的模型都存在很大的泛化误差，但二者的问题是截然不同的。

  Actually, fitting a 5^{th} order polynomial to the data (rightmost figure) did not result in a good model. For example, even though the 5th order polynomial did a very good job predicting y (say, prices of houses) from x (say, living area) for the examples in the training set, we do not expect the model shown to be a good one for predicting the prices of houses not in the training set. In other words, what's has been learned from the training set does not generalize well to other houses. The generalization error of a hypothesis as we defined above is too large. The reader might notice that both the models in the leftmost and the rightmost figures above have large generalization error. However, the problems that the two models suffer from are very different.

偏差–方差权衡
Bias/Variance Tradeoff

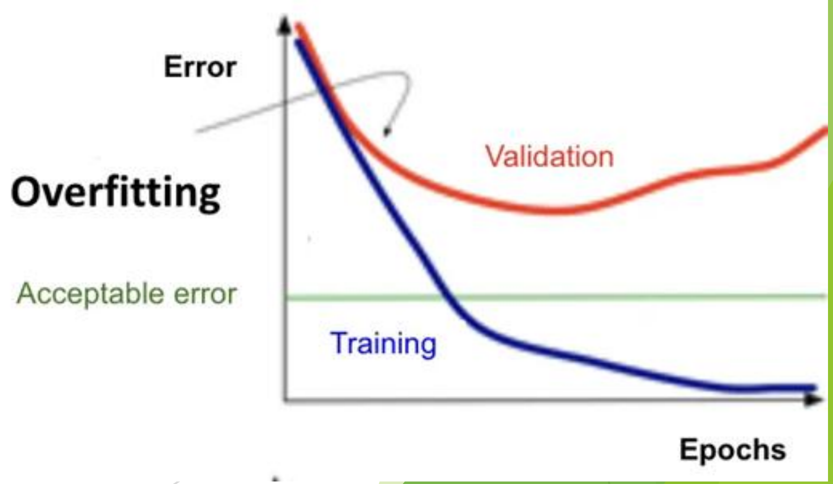
- 如果y与x之间并非线性关系，那么即便用极大规模训练集去拟合线性模型，它依然无法准确捕捉数据中的结构。通俗地说，我们把即便用无限大训练集去拟合，模型依然存在的期望泛化误差定义为模型的偏差。因此在上述问题中，线性模型存在高偏差，可能会对数据产生欠拟合（即无法捕捉数据本身的结构）。如下图所示，此种情况下，无论是测试集还是验证集，其误差皆在可接受的阈值之上。

If the relationship between y and x is not linear, then even if we were fitting a linear model to a very large amount of training data, the linear model would still fail to accurately capture the structure in the data. Informally, we define the bias of a model to be the expected generalization error even if we were to fit it to a very (say, infinitely) large training set. Thus, for the problem above, the linear model suffers from large bias, and may underfit (i.e., fail to capture structure exhibited by) the data. As indicated by the following figure, in this case, the empirical error from training set and validation set are all above the threshold.

偏差–方差权衡
Bias/Variance Tradeoff

- 除了偏差，泛化误差还有第二个组成部分：模型拟合过程的方差。具体来说，当像右图那样拟合 5 阶多项式时，存在很大风险：我们学到的只是小规模有限训练集里碰巧出现的偶然模式，而这些模式并不能反映x与y之间真实、普遍的关系。例如，可能只是训练集中随机出现了几处“略贵于均价”或“略便宜于均价”的房屋。通过拟合这些训练集中的虚假模式，模型同样会出现很大的泛化误差。这种情况，我们称模型具有高方差。

  Apart from bias, there’s a second component to the generalization error, consisting of the variance of a model fitting procedure. Specifically, when fitting a 5th order polynomial as in the rightmost figure, there is a large risk that we’re fitting patterns in the data that happened to be present in our small, finite training set, but that do not reflect the wider pattern of the relationship between x and y. This could be, say, because in the training set we just happened by chance to get a slightly more-expensive-than-average house here, and a slightly less-expensive-than-average house there, and so on. By fitting these 'spurious' patterns in the training set, we might again obtain a model with large generalization error. In this case, we say the model has large variance.



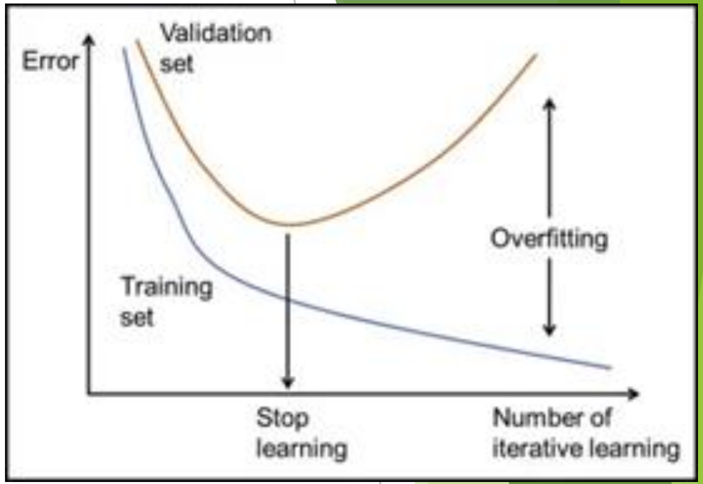
偏差–方差权衡
Bias/Variance Tradeoff

- 通常，偏差与方差之间存在权衡：
  - 如果模型过于简单、参数极少，会出现高偏差、低方差；
  - 如果模型过于复杂、参数极多，则会出现高方差、低偏差。

在上面的例子中，拟合二次函数的效果，要优于一阶多项式（过于简单）和五阶多项式（过于复杂）这两种极端情况。

- Often, there is a trade-off between bias and variance.
  - If the model is too “simple” and has very few parameters, then it may have large bias (but small variance);
  - If it is too “complex” and has very many parameters, then it may suffer from large variance (but have smaller bias).

In the example above, fitting a quadratic function does better than either of the extremes of a first or a fifth order polynomial.



VC置信项
VC Confidence

- 我们不再证明下述不等式以概率$1-\delta$成立：

  We will state the following inequality which holds with probability $1-\delta$ without proof:

$$R[h] \leq R_{\text{emp}}[h] + \sqrt{\frac{v(\log(2l/v)+1)-\log(\delta/4)}{l}}$$

- 我们可以看到，具有最小经验风险的假设h_{emp}^{*}，与任意的$h \in H$，差一项，我们称其为VC边界。我们不再证明，通过最小化权重的范数，可以控制该项的大小，即$\|w\| \propto \text{VC}$界，或$\|\theta\| \propto \text{VC}$界，该项技术称为正则化，有助于提升模型的泛化能力。

  It is manifest that there is an item between the smallest empirical risk h_{emp}^{*} and an arbitrary $h \in H$. It is called the VC bound. We will not prove that minimize the norm of weights can reduce the VC bound, aka., $\|w\| \propto \text{VC bound}$ or $\|\theta\| \propto \text{VC bound}$. This is also called regularization in some textbook, which can boost the performance of generalization.